Entry 660
Hanging recovery

Technical implementation record and single-entry demonstration

RoboTwin / frozen Neotix / Modal T4
Boniface - final rack-hanging phase

Recorded outcome	Evidence
Four recovery stages completed	Approach, seating + settling, guarded release, retreat
Native benchmark success	True at release and at final report
Post-retreat observation	250 physics ticks (1 simulated second)
Public automatic support predicate	Unknown; retained honestly in the JSON report
Scope of claim	One recorded seed-3 entry; repeatability is not established

The saved video and final camera views show the mug released on the peg with the right gripper withdrawn. The implementation deliberately retains **recovery_sequence_completed_support_unverified**: geometric alignment and native success are not a force/contact measurement or an automatic independent support test.

Authoritative successful recording

`hanging_recovery_20261031T104702Z_1793443622777436500`

This document describes the implemented path, failed iterations and the evidence needed to reproduce the development demonstration. It is an engineering record, not certification of general hanging reliability. No new simulator attempt was launched to produce this document.

1. Objective and architecture

The frozen Neotix VLA remains responsible for the recorded left-arm acquisition, table handoff and right-arm pickup. Exactly 660 recorded 14-DoF actions are replayed from reset. A bounded deterministic recovery then replaces the final hanging transition; it does not retrain the VLA, alter the recorded prefix or introduce an arm lock.

Layer	Responsibility
prepare_replay / validate_entry660 / run_entry660	Package the recorded action prefix, enforce explicit unverified opt-in, check hashes and compare the boundary observation.
aperture_frame.py	Fit a visible material plane and intersect reviewed opening rays; report sensitivity, uncertainty and rejected geometry.
grounding_preflight.py	Bounded model proposal/crop validation for the exploratory path; the successful reviewed-aperture demo uses no model calls.
recovery.py	Bind public rack-relative targets, use measured TCP transport, confirm fresh depth, sequence four guarded stages and record hold evidence.
AnalyticSkill / RackSeatReleaseGate	Plan bounded motion, enforce tracking and settling, authorize opening from a current confirmed seat.
Modal launcher / EpisodeVideo	Validate before simulator initialization; preserve reports, RGB-D, event logs and sampled four-camera video.

Public information boundary

Targets use rendered RGB-D, camera calibration, reviewed visible pixels and measured end-effector pose. Simulator object poses and contacts are not used to construct targets. The native benchmark success flag is reported separately. The additional perception_camera is available for grounding/video; the policy payload retains its original three cameras and 14-DoF interface.

Serial and bounded execution

The recovery is development-only and explicitly opt-in. Replay is bounded to 660 actions and 30 minutes. Recovery has a 600-second deadline, five skill calls and 12,000 analytic ticks. Approach/seat each allow at most 3,000 ticks; retreat at most 2,000. The launcher uses single-use containers and no automatic invocation retries; this is not a workspace-wide distributed lock.

2. What failed before success

Recording / experiment	Observed blocker and resulting change
Original 900-action baseline	Right-arm pickup and rack approach reached entry 660; opening around observations 776-784 did not establish a sustained hang. Capture gaps were recovered from the existing volume.
Model proposal preflight	API rejected thinking.type.disabled for the selected model. The bounded wrapper uses adaptive thinking, low effort and SDK retries disabled; low-confidence proposals may use one mapped RGB crop per target.
30 Oct, 09:06 recovery	Approach tracking residual 0.250046 rad exceeded the existing 0.25-rad guard. No execution defect was established and the tracking guard was not widened.
30 Oct, 13:46 recovery	Inconsistent handle semantics: proposals on the same view differed by 42.8 mm / 48.84 degrees. A surface frame was not a reliable aperture mating frame.
Reviewed aperture construction	Thin-surface mixed depth, an opening-edge sample, C-arch coverage and overly pessimistic clearance assumptions blocked preflight. These were resolved explicitly (sections 3-4).
30 Oct, 19:09 recovery	Approach completed but strict post-approach confirmation rejected occluded material samples at observation 1158. Measured TCP was already used; slip was not proven.
31 Oct, 09:19 recovery	-45-degree roll allowed approach and seat. Seat ended at 3.718 mm / 0.0169 rad; a 1-mm demo release gate and unavailable authorization blocked opening.
31 Oct, 10:13 recovery	Nine valid depth matches and one 5.339-mm residual triggered an all-sample veto. Added a capped one-outlier robust rule without changing the 5-mm match threshold.
31 Oct, 10:47 recovery	All four stages completed; final native success true. One second of post-retreat images retained the mug on the peg.

Timestamps identify stored run folders, not a randomized comparison. Multiple tuning iterations on the same entry are development evidence; they are not independent successful trials.

3. Visible aperture reconstruction

A handle-surface frame and an aperture frame have different semantics. Three arbitrary surface landmarks can have tight depth while still yielding the wrong insertion axis or opening center. The final path instead uses a complete, ordered, reviewed opening contour and surrounding solid handle-material samples from observation 660.

Reviewed data and plane fit

Nine contour vertices and twelve material samples are bound to the exact RGB-D archive hash. The plane is fitted to material only using SVD. Contour rays intersect that plane; the opening/background depth itself is not treated as handle material. The polygon must be convex, ordered, non-degenerate and visible. The fitted normal faces the camera; image-up projected into the plane defines its second axis.

Issue	Resolution / measured result
Opening-edge material point (113,49)	Shifted two pixels left to (111,49); the remaining sample set is unchanged. Maximum plane residual: 2.039 -> 1.100 mm. Aperture centroid shift: 1.219 mm; new 3x3 spread: 3.395 mm.
Unrelated nearer pixels	Wide-neighborhood minima (131,46), (127,48), (125,50) falsely rejected solid samples (124,38), (126,40), (128,42): gaps 64.586, 77.028, 87.581 mm.
Surface connectivity refinement	For reviewed aperture samples only: 4-connected depth support, <=10-mm neighbor steps, opening mask exclusion, at least nine supporting pixels. Connected gaps became 20.000, 17.583, 13.380 mm.
Local edge/background bleed	The selected 3x3 patch must still have valid material and <=60-mm spread. Connectivity does not exempt a mixed patch or a sample inside the opening.
Visible C-shaped arch	Require >=150-degree visible angular spread and condition number <=10 rather than full enclosure. Entry 660: 217.089 degrees, condition 3.423.

Sensitivity is not calibrated uncertainty

Maximum residual is 1.0996 mm; tested center sensitivity is 4.3190 mm and normal sensitivity 4.6941 degrees. Tests include four rigid contour shifts, four rigid material shifts and leave-one-out checks. They are not exhaustive independent pixel perturbations or confidence intervals. Systematic depth/calibration error remains unknown. The candidate estimator itself does not authorize motion.

4. Thickness and clearance budgeting

Term	Entry 660 value / meaning
Centroid aperture radius	10.9953 mm, minimum edge distance from the polygon centroid used by the target
Maximum inscribed circle	12.9383 mm; its center differs by 2.670 mm. Diagnostic only; the target is not silently moved to that center.
Peg radius	Nominal 4.000 mm user-authorized prior; not a calibrated radius estimate from thin pixels
Available radial gap	10.9953 - 4.0000 = 6.9953 mm
Visible-plane to handle middle	-4.000 +/- 2.000 mm along the camera-facing plane normal (8 +/- 4 mm full thickness)
Engaged length / axial allowance	30 mm / 30 mm, explicit demo priors
RSS radial sensitivity budget	5.5882 mm including 0.500-mm margin
Axial sensitivity budget	5.2596 mm including 0.500-mm margin

Experimental budget used by the opt-in demo

```
radial_required = sqrt(center_sensitivity^2 + plane_residual^2
 + (engaged_length * sin(normal_sensitivity))^2) + margin
axial_required = hypot(center_sensitivity, thickness_uncertainty / 2) + margin
```

The earlier non-demo budget stacked the analytic endpoint allowance (15 mm / 0.10 rad) linearly with sensitivities; it exceeded the opening radius before a physical fit could be tested. The Entry 660 override uses the explicit RSS sensitivity model above and a tighter measured release gate. It does not certify statistical independence or bound unknown systematic errors.

The 5-mm release position gate is numerically inside the 5.588-mm sensitivity budget and 6.995-mm radial gap, but TCP error is not total aperture error. These values must not be represented as a formal collision-free clearance guarantee. Position/orientation prior fields set to zero mean no additional quantified term, not measured absence of error.

Preflight boundary

Preflight validates the archive binding, public geometry and candidate waypoint bounds before simulator startup. Its IK status is explicitly not_run_requires_simulator_planner. Actual IK/planning is evaluated live through the existing planner; offline geometry must never be described as an offline IK pass.

5. Target transform and wrist roll

```
T_tcp_aperture = inverse(T_world_tcp_measured) * T_world_aperture_middle
T_world_tcp_seat = T_world_peg_conditioned * inverse(T_tcp_aperture)
T_world_tcp_approach = offset_50mm_along_rack_axis(T_world_tcp_seat)
```

The visible surface frame is offset by the explicit thickness prior before forming the measured grasp transform. The rack frame supplies peg-axis conditioning. Candidate roll is sampled at 65 values in the supplied interval; eligible candidates minimize wrist rotation from the measured current pose. The validated Entry 660 choice remains **-0.7853981634 rad (-45 degrees)**.

Why the earlier roll was rejected

At -1.178097 rad (-67.5 degrees), the mug body obscured some predicted handle material after approach. The improved candidate filter keeps the aperture above the visible body and checks predicted sample visibility and distance to the nominal peg segment. These are partial-view heuristics, not a complete mug collision model.

Selected-roll diagnostic	Value
Handle above visible-body mean	13.894 mm; minimum criterion 10 mm
Predicted visible material samples	12; minimum criterion 6
Visible body to peg-segment distance	66.903 mm; minimum criterion 9 mm
Complete collision model	False

Successful TCP targets: world meters, quaternion wxyz

```
rack_approach
xyz: 0.239134203, -0.057101709, 0.885863159
wxyz: 0.478695844, 0.078463569, 0.112711683, 0.867173474

rack_seat
xyz: 0.263192762, -0.038622909, 0.846117472
wxyz: 0.478695844, 0.078463569, 0.112711683, 0.867173474
```

Targets are recomputed from fresh observation identity and measured TCP before each move. No hard-coded start joint state, absolute rack-coordinate command or teleport is used. The numbers above are a report of this entry, not constants to reuse on another episode.

6. Fresh-depth confirmation

The pre-seating check transports reviewed material points by measured post-approach TCP motion: T_delta = T_tcp_now * inverse(T_tcp_entry). The earlier failure at observation 1158 was not caused by using commanded TCP. Six visible points matched within 0.626-2.605 mm; other projected pixels hit nearer occluders, with 37.5-100.5-mm world residuals. These observations did not prove grasp slip.

Classification / guard	Working rule
Per-sample visible match	World-space residual <=5.000 mm; this threshold is unchanged
Nearer occluder	Observed surface >5 mm nearer than predicted: unavailable, not positive evidence
Local depth validity	Reject unavailable or discontinuous selected 3x3 neighborhoods; retain 60-mm spread guard
Distributed visible support	At least six inliers and at least half the reviewed samples, with spatial/plane conditioning
Marginal contradiction policy	At most one contradictory sample, >=10 usable samples, >=9 true 5-mm inliers, median <=5 mm and bounded robust cutoff
Robust cutoff	min(10 mm, max(5.5 mm, median + 3 * 1.4826 * MAD))
Still rejected	Multiple contradictions, large background residuals, degenerate support, stale/wrong-camera evidence

In the 10:13 attempt, one residual was 5.339 mm with nine good matches, so a strict all-sample veto aborted before seating. The revised rule can label that single marginal value trimmed_marginal_outlier; it never counts it as a valid 5-mm match. In the successful run at observation 1181, nine samples were valid matches and **no samples needed trimming**.

What the check establishes

It establishes consistency of a visible material subset with the measured rigid-grasp prediction. Occluded samples provide no confirmation. The frame is transported rather than re-fitted to a changing partial arch. No hidden handle reconstruction or bounded ICP correction is silently applied; grasp force, physical support and calibration accuracy remain unmeasured.

7. Seat, release and retreat handoff

Seating plus bounded settling

After a completed seat plan, recovery holds the existing last planned right-arm joint endpoint with the gripper closed. It allows at most 250 extra ticks, requires at least **25 ticks** and five consecutive measurements within **5.0 mm / 0.05 rad**. It preserves the 0.25-rad tracking guard, inactive-arm hold and deadline. The successful run needed exactly 25 extra ticks.

Authorization is a current measured predicate

RackSeatReleaseGate records only a certain, completed, correct-arm move_to rack_seat result with a perception anchor bound to that command observation. It clears prior authorization for invalid or uncertain results. The aperture recovery path maps unavailable authorization to explicit false with a typed reason; it cannot silently return null. Evaluate compares current measured TCP to the grounded seat, including arm and observation freshness.

The successful demo uses a 5.0-mm position / 0.05-rad orientation gate. Default gates outside this explicit demo path remain unchanged. The original 1-mm demo mismatch was not evidence of failed seating; matching the measured settling criterion and release criterion removed that handoff inconsistency.

Guarded opening

Release uses the existing 200-tick ramp and checks authorization while opening. Alignment loss still stops further opening. Completion means the gripper opened as commanded, not that contact or sustained support has been independently measured. The stage recorded 225 ticks including its own endpoint settling.

Retreat after native success

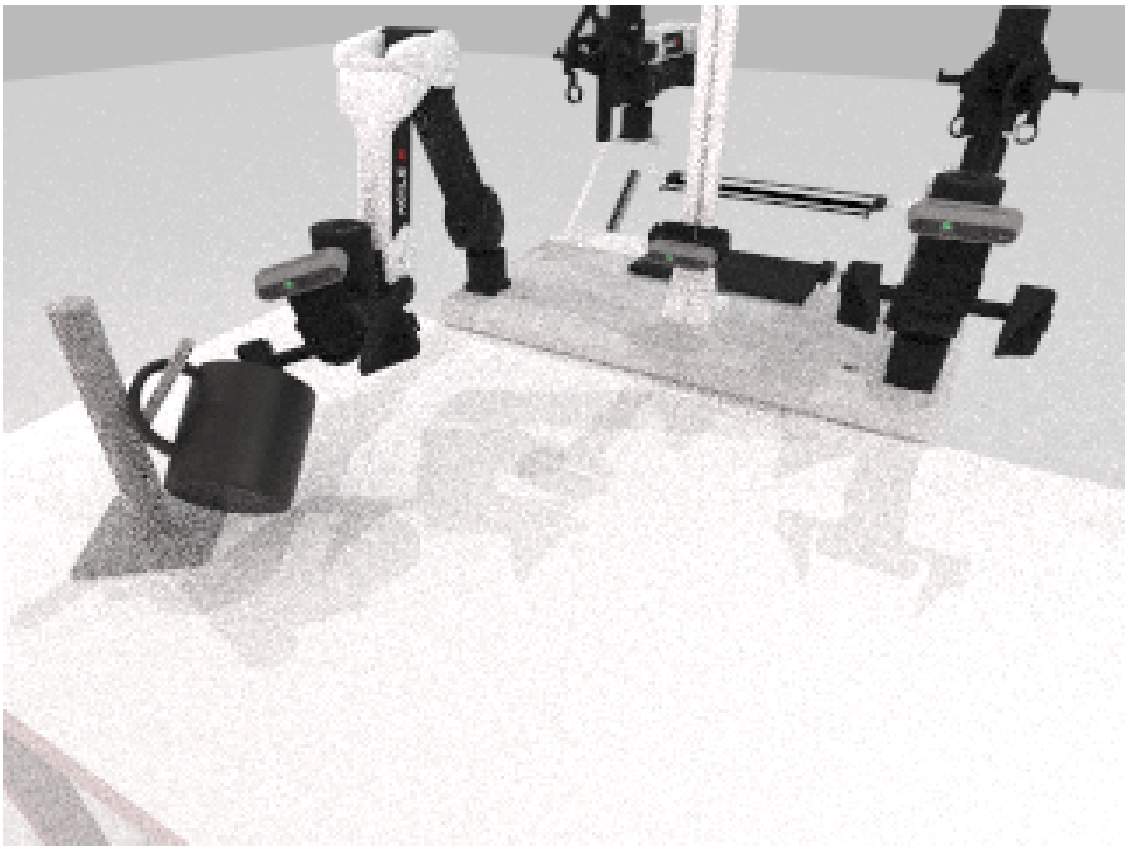
The benchmark can latch success before withdrawal. Only after a certain completed demo release with the right gripper at least 0.99 open does a narrow environment proxy permit the bounded retreat instead of terminating immediately. The underlying native flag remains unchanged and is reported. Planning, workspace, tracking, cancellation and tick limits continue to run.

Retreat begins at measured current TCP and uses the rack-relative approach-to-seat displacement, with the gripper open. A separate 250-tick observation-only hold records the post-retreat state. Approach, seat, release, retreat and support are separate outcomes; success in an earlier stage never overwrites a later failure.

8. Recorded end-to-end result

Stage	Observation range	Ticks	Position error	Orientation error
approach	660 -> 1181	521	2.502 mm	0.005421 rad
seat	1181 -> 1347	166	3.518 mm	0.016272 rad
release	1347 -> 1572	225	-	-
retreat	1572 -> 1712	140	1.911 mm	0.004366 rad

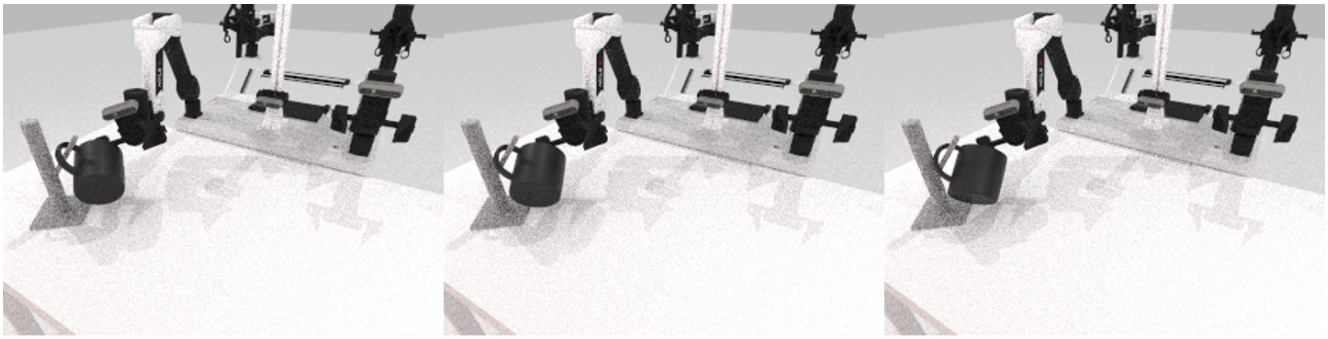
Seat includes 141 motion ticks plus 25 settling ticks. The four stages total 1,052 analytic ticks. The 250-tick post-retreat hold is reported separately, so there are 1,302 additional physics ticks after the 660-action prefix. Observation 1962 is the final saved boundary.



Successful run, observation 1962: perception_camera after guarded release, withdrawal and one simulated second of hold.

Native success: **true**. All four stage stop reasons: **completed**. No policy chunks were inferred during recovery. Source camera images, TCP positions and command state agreed at replay entry (reported errors zero), while capture provenance and full-state restoration remain explicitly unverified.

9. Visual evidence and interpretation



Observation 1712 | 0.0 safter retreat

Observation 1837 | 0.5 safter retreat

Observation 1962 | 1.0 safter retreat

The sampled views at retreat completion, +0.5 s and +1.0 s show the mug retained on the peg with the gripper withdrawn. This supports a released-hang interpretation for this recording. It is not an independent force/contact measurement, a long-duration endurance test or evidence across different entries.

Video timing

episode.mp4 contains 736 frames at 10 fps (73.6 seconds). Recording samples one frame per recorded policy target, one per 25 analytic ticks and boundary frames. Recovery begins at roughly 66 playback seconds. This is sampled playback, not wall-clock or uniform physical-time video. video_frames.jsonl maps every frame to its observation sequence.

Evidence hierarchy

Evidence	What it does and does not establish
Measured pose / depth consistency	Supports geometric execution and visible rigid-grasp consistency; does not prove physical support.
Native benchmark flag	Reports simulator task checker success; not used to fabricate public grounding or a force sensor.
Video / RGB-D visual review	Shows open gripper, separated hand and retained mug for the sampled hold; limited to visible views and duration.
Automatic public_support field	Remains unknown. The report does not promote alignment or opening into an automated supported-hang claim.

10. Verification and cleanup scope

Cleanup preserves the exact validated target math and control settings. An unused estimator import was removed; contour projection remains executed as a visibility guard; a conditional save expression was rewritten as a normal if statement. Existing structured event logs, failure details and launcher status output are operational artifacts, not temporary debug prints.

Focused offline test result

53 tests passed, with 24 subtests passed under pytest after cleanup. This is the focused recovery/gate/release/retreat/preflight/aperture set. Synthetic tests verify behavior and rejection contracts; saved-artifact regressions exercise recorded geometry when those ignored artifacts are available.

```
python -m pytest tests/test_hanging_recovery_runner.py
 tests/test_rack_seat_gate.py tests/test_hanging_release.py
 tests/test_hanging_retreat.py tests/test_hanging_grounding_preflight.py
 tests/test_aperture_frame.py -q
```

Area	Coverage retained
Aperture geometry	Ordered contour, depth separation, visible arch conditioning, sensitivity, hash binding and explicit demo opt-in
Motion	Measured starts, bounded waypoints, tracking, cancellation, settling and execution budgets
Fresh depth	Occlusion, visible subset, single marginal outlier, multiple/gross contradictions and unchanged 5-mm inlier threshold
Release	Certain seat, correct arm and anchor freshness, explicit unavailable rejection and alignment loss during opening
Sequencing	Native success during release, bounded retreat, separate support outcome and final observation hold
Replay / camera / video	Unverified gate, fixed prefix, original policy camera payload and sampled perception-camera recording

Isolated PR snapshot: 190 passed, 5 skipped, 45 subtests passed across the intended test files and existing motion/target/runtime regressions. The snapshot contains HEAD plus only the selected source/test changes, excluding unrelated working edits. Five recorded-artifact checks skip when ignored recordings are absent.

Full Windows suite: 662 passed, 10 failed, 233 subtests passed. Failures are outside the hanging changes: Windows symlink privilege, path separator and os.killpg assumptions, plus missing torch and matplotlib dependencies. These remain disclosed rather than changing unrelated code to make this environment green.

The source PR excludes unrelated SAM 3 service/cache changes, local setup edits, renderer-patch changes, Markdown drafts and raw run artifacts. The generated PDF documents the full development history while the code remains scoped to hanging and the recording/replay tools it requires.

11. Reproduction and evidence inventory

Reproduction requires the ignored recorded-prefix package, reviewed annotation and peg proposal files, and the original archive; these are intentionally not committed. A clean checkout alone cannot reproduce the demonstration without obtaining those artifacts and a compatible simulator/policy asset environment.

```
$env:MODAL_IMAGE_BUILDER_VERSION = "2025.06"
$env:SIM_GPU = "T4"
.\.venv\Scripts\modal.exe run modal_apps/simulator.py::hanging_recovery `
 --model "claude-opus-5-5" --allow-unverified `
 --allow-entry660-demo-override `
 --package <prepared-entry660-package> `
 --reviewed-archive <observation_00000660.npz> `
 --aperture-annotation <reviewed-aperture.json> `
 --peg-annotation <reviewed-peg-proposal.json>
```

Angle-bracket paths identify required existing private artifacts, not shell-ready literal arguments. The explicit model argument remains launcher metadata on this reviewed demo path; no Anthropic inference is made. The non-reviewed model path is separate and is not the demonstrated result.

Stored successful evidence

```
data/runs/hanging_recovery_20261031T104702Z_17934436227774365oo/
 report.json
 preflight/report.json
 recovery_events.jsonl
 video_frames.jsonl
 episode.mp4
 observations/observation_00001962.npz
```

Remote artifacts use the same run identifier in the robotwin-runs Modal volume. Modal run: ap-SEBJP9vRaj6f5WbfgzLSVZ. Keep recordings, checkpoints, weights, credentials and generated raw reports out of Git.

Source archive binding (SHA-256)

```
d510a6fd877d00bae179d51f0c6a94b76304a20c0cccea2313f2ac0deac8e25a
```

Successful report binding (SHA-256)

```
bacb3198e489f0e72427019fd60c01440bcf127d1ce5dfe939ba6db41c5fab16
```

12. Remaining limits and next evaluation

This result completes a development demonstration of bounded hanging recovery from one reproduced entry. It does not establish autonomous public keypoint selection, independently calibrated clearance, force-aware seating or repeatability across episodes.

Limit	Required follow-up before a broader claim
Capture provenance unknown	Record source commit, image/dependency digests, config hashes and policy identity at capture time. Replay agreement is not full simulator-state restoration.
Reviewed pixels	Retain annotation provenance. Current annotations were visually reviewed during development, not independently validated by a separate annotator.
Nominal dimensions	Measure peg radius, handle thickness and systematic camera/depth errors; validate conservative uncertainty budgets.
Partial object visibility	Validate body/handle geometry across views; the current body-distance/visibility heuristic is not a full collision model.
Short hold	Increase declared observation duration and add an independent public stability/support evaluator before claiming long-term retention.
One tuned entry	Freeze implementation, declare held-out entries/seeds and report valid-entry-conditioned and end-to-end rates with denominators.
Execution variability	CuRobo/physics can vary by a tick and endpoint. Retain existing tracking guards and report failures rather than silently widening bounds.

Next experiment after review

Keep this successful Entry 660 configuration fixed. Review and merge the scoped PR first, then authorize one serial held-out-entry test at a time. Preserve acquisition/handoff and the frozen policy interface, save unique ignored evidence paths, and evaluate approach, seating, release, retreat and sustained retention separately.

Documentation basis: local implementation, focused tests and saved reports/observations listed above. No additional live simulator run or grounding API call was performed for this cleanup and documentation task.